

Proforma of information to be collected for the University departments/ADR/ Research sation/ for uploading on University website

1. **Name of the Department/Section :** Department of Agronomy, College of Agriculture, Dapoli, Dist. Ratnagiri (M.S.), India.

: RESEARCH ACHIVEMETS :

Recommendations

Sr. No.	Recommendation				Year
1.	It is recommended to use following weather based regression model for prediction of vegetative flush emergence in Alphonso mango before three or four weeks by using seven days average temperature from second fortnight of September (38 th met week) and onwards in orchards managed by using recommended technology given by Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth, Dapoli under Konkan agro-climatic conditions. ❖ Alphonso mango vegetative flush emergence (3 weeks before) 				

	24 th to 25 th meteorological week (11 to 24 June)	Karjat-7, Karjat-3, Ratnagiri-1 and Phondaghat-1				
5.	It is recommended to sow early, mid late and late rice varieties developed by Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth in Raigad district for obtaining stable yield under varying climatic condition as suggested in following table.					
	Sr. No .	Sowing period	Rice varieties			
			Early	Mid late		Late
	1.	23 rd meteorological week (04 to 10 June)	-	Ratnagiri-6, Ratnagiri-7, Karjat-5 and Karjat-9		Ratnagiri-8
	2.	24 th meteorological week (11 to 17 June)	Karjat-7, Karjat-3 and Phondaghat-1	-	-	
6.	It is recommended that, to use 1:3 row proportion for obtaining higher net returns and B:C Ratio under Mustard + Cowpea intercropping system during <i>rabi-hot</i> weather season of Konkan region.				2022	
7.	In North Konkan coastal zone of Maharashtra, fodder maize + berseem inter-cropping in 2:1 ratio is recommended for obtaining significant higher proteinous yield of fodder and net returns.				2022	
8.	It is recommended to grow groundnut in Konkan during Rabi –Summer season under paddy straw mulch and be irrigated daily through drip irrigation with total irrigation depth of 371 ha mm and fertigated with 100% RDF (25kg N and 50 kg P ₂ O ₅ ha ⁻¹) through water soluble fertilizers in five equal splits of six days interval to get maximum yield and more economic returns and water saving.				2022	
	DAS	Source of Fertilizer	Splits	Quantity of Fertilizers (Kg/ha)		
	6	1)12:61:0	16:39	22.98		
		2) Urea	6:59			
	12	3)12:61:0	16:39	22.98		
		4) Urea	6:59			
	18	5) 12:61:0	16:39	22.98		
		6) Urea	6:59			
	24	7) 12:61:0	16:39	22.98		
		8) Urea	6:59			
	30	9) 12:61:0	16:39	22.98		
		10) Urea	6:59			

9.	It is recommended to grow dibbled rice-sweet corn-green gram cropping system on raised bed or flat bed with silver-black polythene mulch (30 micron) to obtain higher yield and economic benefit by using practices given in following table.																
	<table> <tr> <th>Particulars</th><th>Raised bed</th><th>Flat bed</th></tr> <tr> <td>1. Bed size</td><td>1.0 m. top and 1.20 m. bottom breadth, 20.25 m. in length and 8-10 cm in height</td><td>4.60 m. in breadth and 20.25 m. in length</td></tr> <tr> <td>2. Spreading of Polythene mulch</td><td>Spread single strip of silver-black polythene mulch on the bed</td><td>Seal the four strip of silver-black polythene mulch together and spread it on the bed</td></tr> <tr> <td>3. Spacing</td><td colspan="2"> Rice : 20 x 15 cm. Sweet corn : 40 x 30 cm. Green gram : 20 x 15 cm. </td></tr> <tr> <td>4. Fertilizer dose</td><td colspan="2"> Rice : 100: 50 : 50 NPK kg/ha. Sweet corn : 200 : 60 : 60 NPK kg/ha. Green gram : 25 : 50 : 0 NPK kg/ha. </td></tr> </table>	Particulars	Raised bed	Flat bed	1. Bed size	1.0 m. top and 1.20 m. bottom breadth, 20.25 m. in length and 8-10 cm in height	4.60 m. in breadth and 20.25 m. in length	2. Spreading of Polythene mulch	Spread single strip of silver-black polythene mulch on the bed	Seal the four strip of silver-black polythene mulch together and spread it on the bed	3. Spacing	Rice : 20 x 15 cm. Sweet corn : 40 x 30 cm. Green gram : 20 x 15 cm.		4. Fertilizer dose	Rice : 100: 50 : 50 NPK kg/ha. Sweet corn : 200 : 60 : 60 NPK kg/ha. Green gram : 25 : 50 : 0 NPK kg/ha.		2022
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10.	Spraying of pre emergence herbicide Oxadiargyl 80 % WP @ 100 g a.i.ha ⁻¹ (125 g per ha market product) at 2-3 DAS <i>fb</i> Metasulfuron- methyl + chlorimuron – ethyl @ 4 g a.i. ha ⁻¹ at (125 ml per ha market product) 25 DAS or Oxadiargyl 80 % WP @ 100 g a.i.ha ⁻¹ (125 g per ha market product) <i>fb</i> 1HW at 30 DAS in dry direct seeded rice during <i>kharif</i> season for effective control of weeds and for obtaining higher yield and net returns under Konkan region is recommended.	2022															
11.	It is recommended to grow elephant foot yam-okra crops in sequence under lateritic soils of Konkan region with drip irrigation system and irrigation should be scheduled on alternate day at 100 % PE (Total water applied to elephant foot yam 117.54 mm and okra crop 224.17 mm) and 125 % RDF (Elephant foot yam-FYM @ 10 t ha ⁻¹ + 100:75:100 kg N:P ₂ O ₅ :K ₂ O ha ⁻¹ and Okra-FYM @ 10 t ha ⁻¹ + 125:65:65 kg N:P ₂ O ₅ :K ₂ O ha ⁻¹) through straight fertilizers for obtaining higher system production and economic returns.	2021															
12.	It is recommended that, to get higher yield and economic returns from rice based cropping systems, Rice crop be grown during <i>Kharif</i> season followed by Bottle gourd or Brinjal or Groundnut or Okra during <i>Rabi</i> season under North Konkan Coastal Region.	2021															
13.	It is recommended to apply 170 kg ha ⁻¹ Konkan Annapurna Briquette + 5 ton FYM (25% RDN) to <i>kharif</i> rice and 75% RDF (150: 45: 45 kg NPK ha ⁻¹) to rabi sweet corn for obtaining optimum yield and economic returns in rice- sweet corn cropping system in South Konkan Coastal Zone.	2021															
14.	Spinach: Nutrient management It is recommended that under coastal saline soil of North <i>Konkan</i> region to obtain maximum yield and highest monetary returns, spinach variety <i>Pusa Harit</i> be cultivated	2019															

	with application of nitrogen @ 75 kg ha ⁻¹ and 50 kg P ₂ O ₅ ha ⁻¹ (Khar Land Research Station, Panvel)	
15.	In North <i>Konkan</i> coastal zone of Maharashtra, rice-fodder maize and rice-berseem food-fodder cropping sequences are recommended for obtaining higher yield and economic returns. (Agricultural Research Station, Palghar)	2019
16.	Rabi- summer Groundnut: Nutrient management It is recommended to apply FYM @ 5 t ha ⁻¹ and phosphorus @ 50 kg ha ⁻¹ fertilizer dose at the time of sowing and seed dressing with phosphorus solubilizing bacteria (DGRC 2) @ 25 g kg ⁻¹ for the maximum dry pod yield and net monetary returns in <i>rabi</i> hot weather groundnut in <i>Konkan</i> region. (A.S. Kambale, B.D. Waghmode, V.V. Sagvekar and P.D. Chendage, V.C. Navhale and N.G. Sonone)	2019
17.	Kharif groundnut: Use of Paclobutrazol Foliar spraying of paclobutrazol @ 100 ppm at 30 and 50 days after emergence is recommended for obtaining maximum pod yield and monetary returns from <i>kharif</i> cultivation of groundnut cultivar TKG Bold in lateritic soils of <i>Konkan</i> region. (A.S. Kambale, B.D. Waghmode, V.V. Sagvekar and P.D. Chendage, V.C. Navhale and N.G. Sonone)	2019
18.	Rabi- summer groundnut: Management practices Groundnut variety <i>Konkan Bhuratna</i> be sown with spacing of 30 cm x 10 cm and application of 125% RDF (31.25 kg N and 62.5 kg P ₂ O ₅) ha ⁻¹ along with FYM @ 5 t ha ⁻¹ is recommended for obtaining maximum pod yield and monetary returns under lateritic soils of <i>Konkan</i> region. (A.S. Kambale, B.D. Waghmode, V.V. Sagvekar and P.D. Chendage, V.C. Navhale and N.G. Sonone)	2019
19.	Rice: Sowing period and age of seedling It is recommended to grow rice hybrid Sahyadri 3 in <i>kharif</i> season by sowing the nursery during 23 rd meteorological week (4 June to 10 June) and transplanting 15 days old seedlings for obtaining higher yield and net returns under south <i>Konkan</i> condition. (Dr. M.S. Jadhav, Dr. U.V. Mahadkar, Dr. S.A. Chavan, V.A. Rajemahadik, V.N. Shetye, Dr. S.B. Gangawane, V.M. Kanade, Dr. A.P. Chavan, Dr. V.G. More, Dr. D.N. Jagtap and Dr. S.S. Pinjari)	2018
20.	It is recommended that rice- groundnut, rice- sweet corn and rice- dolichos bean system be grown under organic package of practices to get higher yield and economic returns from rice based cropping system. (RARS., Karjat)	2018

21.	Sugarcane: Planting material and media For obtaining higher yield from sugarcane in South <i>Konkan</i> region, it is recommended to use the seedlings of single bud set grown in the media comprised of coco-peat and vermi-compost in 1:1 proportion along with <i>acetobacter</i> culture @ 5.00 g kg⁻¹. (Dr. M.S. Jadhav, Dr. S.B. Gangawane, Dr. V.N. Shetye, Shri. V.A. Rajemahadik, Dr. S.A. Chavan and Dr. U.V. Mahadkar)	2018
22.	Establishment techniques in rice In lateritic soil of south <i>Konkan</i> coastal zone it is recommended to grow direct seeded rice by adopting conservation tillage on flat bed system along with the use of <i>Konkan Annapurna</i> briquettes in combination with soil application of zinc sulphate and copper sulphate @ 175, 25 and 5 kg ha⁻¹, respectively for obtaining higher yield and net returns. (Dr. U.V. Mahadkar, M.S. Jadhav, V.A. Rajemahadik, V.N. Shetye, Dr. S.A. Chavan, V.G. Chavan, Dr. H.M. Patil, Dr. S.S. Pinjari and Dr. D.N. Jagtap)	2017
23.	IFS model for Kharland The different farming components such as crops [rice (0.50 ha), vegetables (0.27 ha)], Horticulture crops [Coconut (0.15 ha), Sapota (0.03 ha) and spices (0.01 ha) on bund], livestock [Fish pond (0.2035 ha) and Poultry (0.0035 ha)] and complementary [Vermicompost (0.0040 ha), Kitchen garden (0.0028 ha)] are recommended in north <i>Konkan</i> Coastal saline soils. B: C ratio increases if size of pond is increased in IFS. (Khar Land Research Station, Panvel)	2017
24.	Rice- rice cropping system: Nutrient management In North <i>Konkan</i> Coastal Zone of Maharashtra, Rice-Rice cropping system be supplied with recommended dose of NPK along with zinc (120:50:50:6 kg ha⁻¹) to <i>Kharif</i> rice (hybrid variety) rice and recommended dose of NPK (120:50:50 kg ha⁻¹) to <i>Rabi</i>/ Summer (improved variety) rice for obtaining higher yield and economic returns. (RARS., Karjat)	2017
25.	Rice- sweet corn cropping system: Nutrient management To get higher yield and economic returns by sustaining soil fertility and productivity, application of 50 per cent RDF as inorganics and 50 per cent RDN through FYM to Rice – Sweet corn cropping system is recommended. (RARS., Karjat)	2017
26.	Rice- brinjal or Rice- sweet corn cropping system: Resource conservation To get higher yield and economic returns, it is recommended to grow Rice – Brinjal or Rice – Sweet corn system under minimum tillage along with the application of 125 per cent RDF to both the systems and application of rice straw mulch @ 3 t ha⁻¹ to Brinjal and	2017

	Sweet corn for resource conservation.				
	(RARS., Karjat)				
27.	IFS Model (1 ha)				2017
Integrated Farming System Model is recommended for small and marginal farmer of North Konkan Coastal Zone of Maharashtra (Area- 1.00 ha)					
I. Cropping Systems					
Kharif season		Rabi season			
Crop	Area (ha)	Crop	Area (ha)		
Rice	0.20	Brinjal	0.10		
		Water melon	0.10		
Finger Millet	0.05	Cowpea	0.05		
Ground nut	0.10	Field Bean	0.10		
Cucumber	0.10	Sweet corn	0.10		
Fodder crop-Napier Bajara Hybrid (Perennial)	0.05	Fodder crop-Napier Bajara Hybrid (Perennial)	0.05		
Total I	0.50	Total	0.50		
II. Horticulture					
1. Mango		Ratna, Keshar and Alphonso	0.20		
2. Aonla		Krishna, Kanchan and Chakayya	0.05		
3. Sapota		Kali patti	0.05		
4. Coconut + Intercrops i. Black pepper ii. Cinnamon iii. Nutmeg		Pratap Panniyur-1 Konkan Tej Konkan Sugandha	0.05		
5. Nursery- Mango grafts Sapota grafts		Ratna, Keshar and Alphonso Kali patti	0.05		
		Total II	0.40		
III. Livestock					
Dairy Animals 3 cows		2 Crossbred Jursey + 1 Local	35.75 m ²		
Goat unit (10 F + 2 M)		Konkan Kanyal	35.75 m ²		
Poultry 3 to 4 batches/year (150 to 200 birds/batch)		Giriraj and Kadaknath	35.75 m ²		
		Total III	107.25 m ²		

	IV. Complementary enterprise					
	Vermicompost unit	Eisenia fetida	18.00 m ²			
		Total IV	18.00 m ²			
	V. Land for other uses					
	Stores, threshing yard, operational area, roads, bunds, etc.		874.75 m ²			
	Total V		874.75 m ²			
	Grand Total (I+II+III+IV+V)		1.00 ha			
28.	Rice based cropping system: Sweet corn and brinjal					
	In south <i>Konkan</i> Coastal Zone of Maharashtra, Rice- sweet corn and Rice Brinjal cropping systems are recommended for obtaining higher yield and economic returns			2017		
	(ARS., Phondaghat)					
29.	Summer Groundnut: Mulching, hydrogel and nutrient management					
	For obtaining higher production, profit and better water use efficiency with saving of 25% water under lateritic soils of <i>Konkan</i> in groundnut, application of hydrogel @ 5.0 kg ha ⁻¹ and use of integrated nutrient management (7.5 t ha ⁻¹ FYM + RDF 25:50:00 NPK kg ha ⁻¹) is recommended.			2017		
	(A.S. Kambale, B.D. Waghmode, V.V. Sagvekar and V.C. Navhale)					
30.	Rabi-Summer groundnut: Plant population and nutrient management					
	Application of 125% RDF (31.25 kg N + 62.5 kg P ₂ O ₅ ha ⁻¹) along with recommended plant spacing (30 cm x 15 cm) to <i>rabi</i> summer groundnut is recommended to obtain higher production and profitability under lateritic soils of <i>Konkan</i>			2017		
	(A.S. Kambale, B.D. Waghmode, V.V. Sagvekar and V.C. Navhale)					
31.	Organic: Rice based cropping systems					
	In North <i>Konkan</i> Coastal Zone of Maharashtra, it is recommended that rice – groundnut, rice – sweet corn and rice – <i>Dolichos</i> bean systems be grown under organic nutrient management using different organic sources as detailed below, to get higher yield and economic returns.					
	Source	Kharif rice	Rabi crops			
			Groundnut	Sweet corn	Dolichos bean	2017
	FYM (t ha ⁻¹)	5.0	1.5	6.0	4.0	
	<i>Glyricidia</i> green leaves (t ha ⁻¹)	7.5	--	--	--	
	Neem cake (t ha ⁻¹)	0.5	0.150	0.6	0.4	
	Rice straw (t ha ⁻¹)	4	--	--	--	
	Vermicompost (t ha ⁻¹)	--	0.5	2.0	1.3	
	Two sprays of cow urine and Vermiwash 10 %	--	--	50	--	

	each at 30 and 60 DAS (lit ha ⁻¹)															
32.	<i>Suru</i> Sugarcane: Planting geometry and intercropping system In <i>Konkan</i> region it is recommended that, for obtaining higher yield and net returns, <i>suru</i> sugarcane be planted in paired rows at 60 cm X 60 cm – 120 cm and intercropped with two rows of sweet corn at 45 cm spacing between paired row. (Dr. M.S. Jadhav, Dr. S.B. Gangawane, Dr. V.N. Shetye, Shri. V.A. Rajemahadik, Dr. S.A. Chavan and Dr. U.V. Mahadkar)					2017										
33.	Sugarcane: Planting layout and nutrient management In lateritic soils of <i>Konkan</i> region for obtaining higher yield and net returns from sugarcane, it should be grown by paired row planting on ridges and furrows with drip irrigation by using single bud settling raised in soil + FYM in 1: 1 proportion and should be fertilized with recommended dose of fertilizers (250: 125: 125 NPK ha ⁻¹) through straight fertilizers. (Dr. M.S. Jadhav, Dr. S.B. Gangawane, Shri. V.A. Rajemahadik, Dr. S.A. Chavan and Dr. U.V. Mahadkar)					2017										
34.	Okra: Nutrient and irrigation management It is recommended to grow okra in Red Ferrogeous soils of <i>Konkan</i> region at a spacing of 120-45 cm x 15 cm in paired row under drip irrigation with plastic mulch and be irrigated daily by following the given schedule with RDF (100: 50: 50) through WSF in seven equal weekly splits through drip irrigation to achieve higher productivity and economic returns. <table><tr><td>Crop Period (weeks)</td><td>Water application (lit/m length)</td></tr><tr><td>1 to 5</td><td>52</td></tr><tr><td>6 to 9</td><td>50</td></tr><tr><td>10 to 13</td><td>75</td></tr><tr><td>14 to 17</td><td>47</td></tr></table>					Crop Period (weeks)	Water application (lit/m length)	1 to 5	52	6 to 9	50	10 to 13	75	14 to 17	47	2017
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1 to 5	52															
6 to 9	50															
10 to 13	75															
14 to 17	47															
35.	It is recommended that in <i>Konkan</i> region, for effective weed control and higher returns form <i>Kharif</i> drilled rice, pre emergence application of pendimethalin @ 1.00 kg./ha (30 EC) followed by one hand weeding at 25 DAS. (Dr. S.B. Gangawane, M.J. Mane, Dr. S.S. Pinjari and V.M. Kanade)					2016										
36.	Rice- groundnut cropping system: Weed management In the <i>Konkan</i> region, for obtaining higher yield, net returns and effective weed control in the rice-groundnut cropping system, incorporation of green manure (<i>Sesbania rostrata</i>) and application of herbicide namely pretilachlor (PE) @ 0.75 kg/ha 3 to 7 DAT to <i>kharif</i> rice and pendimethalin (PE) @ 1.00 kg/ha 2 to 3 DAS to <i>rabi</i> groundnut is recommended.					2016										

	(Dr. V.B. Newase, M.J. Mane, Y.R. Govekar, Shri. V.M. Kande and Dr. S.B. Gangawane)	
37.	<i>Rabi</i> Groundnut- <i>kharif</i> rice cropping system: Nutrient management It is recommended to apply 25 kg N + 75 kg P ₂ O ₅ ha ⁻¹ to groundnut and 75% RDF (75 kg N + 37.50 kg P ₂ O ₅ + 37.50 kg K ₂ O ha ⁻¹) to rice for getting higher yield and economic returns from <i>rabi</i> groundnut- <i>kharif</i> rice system under South <i>Konkan</i> Coastal conditions. (V.V. Sagvekar, B.D. Waghmode, A.S. Kambale V.C. Navhale)	2016
38.	<i>Kharif</i> Groundnut: Planting technique In <i>Konkan</i> region, it is recommended to grow <i>kharif</i> groundnut on Broad Bed and Furrow (BBF) at 80 - 20 cm using 7 micron 44 kg ha ⁻¹ transparent polythene mulch for getting higher yield and economic returns. (A.S. Kambale, B.D. Waghmode, V.V. Sagvekar and V.C. Navhale)	2016
39.	To get higher yield and economic returns from direct seeded <i>kharif</i> rice variety <i>Panvel 3</i> be grown in North <i>Konkan</i> coastal saline soils having 2.5 to 8.5 d Sm ⁻¹ EC with application of 100% recommended fertilizer dose (100:50:50 N, P ₂ O ₅ , K ₂ O kg ha ⁻¹) (Khar Land Research Station, Panvel)	2016
40.	Sweet corn: Nutrient management In lateritic soil of <i>Konkan</i> region, it is recommended to grow sweet corn during <i>rabi</i> season under drip irrigations with application of soil test based major fertilizers along with micronutrients viz. Cu, Zn., B and Mn and amelioration with 50% lime requirement for obtaining higher yield and net return and B: C ratio (Dr. U.V. Mahadkar, V.N. Shetye, V.A. Rajemahadik, Dr. R.T. Thokal, Dr. A.S. Kamble, M.S. Jadhav, Dr. S.A. Chavan and V. M. Kanade)	2016
41.	Direct seeded :Seed rate and nutrient management 12. In coastal saline soil of North <i>Konkan</i> region to obtain higher grain yield with high net profit from <i>Panvel 1</i> rice variety under direct seeded method, crop seeded @ 100 kg ha ⁻¹ with application of nitrogen dose @ 100 kg ha ⁻¹ along with basal dose of P ₂ O ₅ and K ₂ O is recommended (Khar Land Research Station, Panvel)	2015
42.	Cowpea: Zero tillage, irrigation and nutrient management It is recommended to grow cowpea under zero tilled condition during <i>rabi</i> season and two irrigations (at branching and pod filling stage) along with 100% recommended dose of fertilizer (25: 50: 00 N & P kg ha ⁻¹) should be applied below seed for obtaining higher yield and profitability. (Dr. U.V. Mahadkar, Dr. V.N. Shetye, V.A. Rajemahadik, Dr. S.A. Chavan, Dr. L.S. Chavan, V.M. Kanade, Dr. S.B. Gangawane, M.S. Jadhav)	2015

43.	Mango: Weather forecast model Under recommended package of practices given by Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth, Dapoli; the following weather parameters based model is recommended for one week before prediction of flowering during the period of 1st December to 15th January in Alphonso mango under South <i>Konkan</i> coastal agro climatic conditions. Flowering (forecast 1 week early) = - 91.91 + 10.79 Tmax + 6.05 Tmini – 3.40 RH-I + 0.86 RH-II – 5.04 BSS – 3.48 Rainfall – 12.62 Rainy days $R^2=0.79^{**}$ (Dr. V.G. Chavan, Dr. S.T. Thorat, Dr. S.B. Gangawane, Dr. V.N. Shetye, Shri. V.A. Rajemahadik, Shri. V.M. Kanade, Dr. S.A. Chavan and Dr. U.V. Mahadkar)	2015
44.	Mango: Weather forecast model Under standard package of practices as per recommendation of Dr. Balasaheb Sawant Konkan Krishi Vidyapeeth, Dapoli, the following weather parameter based prediction model is recommended for three week before prediction of emergence of vegetative flush during the period of last week of September to 1st week of November in Alphonso mango under South <i>Konkan</i> coastal agro climatic conditions. Alphonso mango vegetative flush emergence (3 weeks before) = 49.47 + 0.44Tmax - 0.18 RH-II - 0.03 Rainfall – 1.61 Evaporation $R^2 = 0.94^{**}$ (Dr. V.G. Chavan, Dr. S.T. Thorat, Dr. S.B. Gangawane, Dr. V.N. Shetye, Shri. V.A. Rajemahadik, Shri. V.M. Kanade, Dr. S.A. Chavan and Dr. U.V. Mahadkar)	2015
45.	Kharif Groundnut: Resource management It is recommended to give first preference to fertilizer management followed by weed management and plant protection measures, respectively under economical constraints for obtaining higher productivity and profit from Kharif groundnut under lateritic soils of <i>Konkan</i> (V.V. Sagvekar, B.D. Waghmode, S.A. Chavan, B.R. Salvi, U.V. Mahadkar and K.E. Lawande)	2014
46.	In <i>Konkan</i> region, for obtaining higher yield and net returns from direct seeded Kharif rice, it is recommended conventional tillage be followed and zero tillage be adopted for succeeding rabi lablab bean (<i>purpureus</i>) in combination with two hand weedings to each crop at 20 & 40 DAS. If there is labour scarcity for hand weeding, pre-emergence application of oxadiargyl @ 0.12 kg/ha for both the crops is recommended. (Prof. R.R. Khadase, M.J. Mane, Dr. V.B. Newase, Dr. L.G. Pawar, Dr. S.T. Thorat)	2013
47.	For effective and economical weed management in direct seeded drilled rice and intern its higher productivity under conditions of south <i>Konkan</i> coastal zone, pre emergence application of oxyfluorfen @ 300 g/ha integrated with PoE application of 2,4-D @ 500 g/ha or hand weeding twice (20 & 40 DAS) is recommended. (M.J. Mane and Dr. V.B. Newase)	2013

48.	Rabi Groundnut: Nutrient management For obtaining higher productivity and profit from <i>rabi</i> summer groundnut in Lateritic soils of <i>Konkan</i> it is recommended to apply 100 % RDF (25 Kg N + 50 Kg P₂O₅) at the time of sowing and 50 % RDF (12.5 Kg N + 25 Kg P₂O₅) as top dressing at one month after sowing. (V.V. Sagvekar, B.D. Waghmode, V.N. Shetye, S.A. Chavan, U.V. Mahadkar)	2013
49.	Kharif Groundnut: Micronutrient management It is recommended to grow groundnut with the soil application of 20 kg ZnSO₄ ha⁻¹ along with recommended dose of fertilizer (25 kg N + 50 kg P₂O₅) for obtaining maximum yield with higher net returns during Kharif season under South Konkan conditions. (V.V. Sagvekar, B.D. Waghmode, V.N. Shetye, S.A. Chavan, U.V. Mahadkar)	2013
50.	Groundnut: Weed management For effective and profitable weed control in rabbi summer groundnut under south Konkan conditions, pre emergence application of Pendimethalin @ 1 kg ha⁻¹ combined with one hand weeding at 30-35 days after sowing is recommended. If hand weeding is not possible, pre emergence application of Pendimethalin @ 1 Kg ha⁻¹ be combined with post emergence application of either Quizalofop ethyl @ 50 g ha⁻¹ or Imazethapyr @ 75 g ha⁻¹ (V.V. Sagvekar, B.D. Waghmode, V.N. Shetye, S.A. Chavan, U.V. Mahadkar)	2013
51.	Sweet corn: Irrigation and nutrient management It is recommended to grow sweet corn crop (Variety- Sugar 75) in lateritic soil of <i>Konkan</i> region under inline drip irrigation system and irrigation should be scheduled on alternate day at 7.4 to 19.0 lit plant⁻¹ from January to April (total water 46.3 ha-cm) with 80% of recommended dose RDF (160: 48: 48 kg ha⁻¹, N: P: K) through WSF to get higher production, better quality and benefit. (Dr. R.T. Thokal and Dr. T.N. Thorat)	2013
52.	Green chilli: Irrigation and nutrient management It is recommended that, in lateritic soil of <i>Konkan</i> region, the green Chilli (<i>Cv. Konkan kirti</i>) crop should be grown under micro-sprinkler irrigation system and irrigation should be scheduled on alternate day with 100% PE (total water 60 cm) should be applied with recommended dose (150: 50: 50, N: P: K) of fertilizer to get maximum production. (Dr. R.T. Thokal and Dr. T.N. Thorat)	2013
53.	Banana: Micro irrigation It is recommended to grow banana crop (<i>cv Safed velchi</i>) in lateritic soil of <i>Konkan</i> region, with microjet irrigation and be irrigated on alternate day 13 to 15 lit. plant⁻¹ during October to January and 18.00 to 21.00 lit. plant⁻¹ during February to onset of monsoon. (Shri. V.A.	2013

	Rajemahadik, Dr. V.N. Shetye, Dr. S.A. Chavan, Dr. U.V. Mahadkar, Dr. R.T. Thokal, Shri. M.S. Jadhav, Shri. V.G. Chavan, Shri. V.M. Kanade and Dr. S.B. Gangawane)	
54.	Rice- rice cropping system: Nutrient management It is recommended that for yield maximization in hybrid rice – hybrid rice system, the crop be fertilized @ 150: 100: 150: 0.8: 10: 6 kg N, P ₂ O ₅ , K ₂ O, B, Fe, Zn ha ⁻¹ during <i>kharif</i> and @ 150:100:150 kg N, P ₂ O ₅ , K ₂ O ha ⁻¹ during <i>rabi</i> season under North Konkan Coastal Zone of Maharashtra. (RARS., Karjat)	2012
55.	Lablab bean: Weed management It is recommended that for effective control of <i>Cuscuta</i> on lablab bean, the field be ploughed and <i>Pendimethalin</i> @1.0 kg ha ⁻¹ as pre- emergence with sand mix be applied to obtain higher yield and net returns. (Dr. L.G. Pawar, Dr. V.B. Nevase and M.J. Mane)	2012
56.	White Onion: Nutrient and weed management It is recommended that onion local cultivar <i>Alibag White</i> be fertilized with 150 kg N + 75 kg P ₂ O ₅ + 25 kg K ₂ O ha ⁻¹ and for effective weed control oxyfluorfen @ 0.176 kg a.i. ha ⁻¹ be applied 4 days after planting followed by one hand weeding at 50 days after planting. Under the scarcity of labourers, the crop be supplied with the same fertilizer dose and for effective weed control oxyfluorfen @ 0.176 kg a.i. ha ⁻¹ be applied 4 days after planting to get higher yield and net returns under the condition north Konkan coastal zone. (Prof. V.N. Khade and M.J. Mane)	2012
57.	Rice- brinjal cropping system In North <i>Konkan</i> Coastal Zone, it is recommended to follow Rice– Brinjal Cropping sequence as most profitable proposition. (RARS., Karjat)	2011
58.	1. Under Konkan conditions, <i>Kharif</i> rice be established in uplands by system of rice intensification (SRI) at 25 cm X 20 cm spacing and for effective weed management hoeing by rotary weeder be carried out for obtaining higher yield and net returns. (Dr. L.G. Pawar and Dr. V.B. Newase)	2011
59.	Under conditions of <i>Konkan</i> , direct seeded dibbled rice be sown in uplands before onset of monsoon and for effective weed management pretilachlor-with safener (50 EC) be applied as pre emergence 0.5 kg ha ⁻¹ for obtaining higher yield and net returns. (Dr. L.G. Pawar and Dr. V.B. Newase)	2011
60.	Banana: Intercropping and irrigation management	2011

	It is recommended that, in lateritic soil of <i>Konkan</i> region, the Banana (cv. Grand Naine) should be grown as inter crop in Arecanut plantation for first three years under drip irrigation system and water should be applied @10-12 lit day⁻¹ plant⁻¹ from November to January and 15-18 lit day⁻¹ plant⁻¹ from February to May to get additional benefit from inter cropping. (Dr. R.T. Thokal and Dr. T.N. Thorat)	
61.	1. Paired row planted dibbled hybrid rice under upland conditions may be grown (15x15-30 cm) in a single way skipping pattern and the crop may be manured 7 WAS either with 7.5 tons <i>Glyricidia</i> green leaves or in situ grown <i>S. rostrata</i> crop in skipped rows @ 6 t ha⁻¹. For effective weed management in such a crop hoeing with <i>Japanese</i> hoe 2, 4 and 7 WAS should be integrated with a manual weeding 6 WAS. (Dr. L.G. Pawar and Dr. S.T. Thorat)	2010
62.	Rice- rice cropping system: Nutrient management In North Konkan Coastal Zone of Maharashtra to improve soil fertility and sustain productivity of rice under rice- rice cropping systems, 50 per cent of RFD (50: 25: 25 kg N, P₂O₅ and K₂O ha⁻¹) as inorganics should be integrated with rest 50 kg N of RFD in the form of either FYM (10 t ha⁻¹) of <i>Glyricidia</i> green leaves (10 t ha⁻¹) during <i>Kharif</i> and 100 per cent RFD as inorganics (120: 50: 50 kg N, P₂O₅ and K₂O ha⁻¹) should be applied during rabi-hot weather season. (RARS., Karjat)	2010
63.	Weed control in rice For effective and economical weed control in rice crop, in case of rice-rice cropping system in medium black soils of Raigad district, pre-emergence application of pretilachlor @ kg ha⁻¹ to <i>kharif</i> rice under drained condition be followed. For <i>rabi</i> hot weather rice, pre-emergence application of butachlor @ 1.25 kg ha⁻¹ plus combination product of metsulfuron methyl 10% + chlorimuron ethyl 10% WP i.e. Almix @ g.a.i. ha⁻¹ 3 days after transplanting be followed. However, whenever field draining is not practically feasible two hand weddings to <i>kharif</i> rice at 25 and 45 days after transplanting and one hand weeding to <i>rabi</i> rice at 40 days after transplanting be followed. (Dr. L.G. Pawar and M.J. Mane)	2008
64.	Organic farming: Varieties <i>Sahyadri-3, Sahyadri-4, Sahyadri-5, Karjat-3, Karjat-5, Karjat-8</i> and <i>Ratnagiri-3</i> rice hybrids/ varieties are recommended to grow under organic package for getting higher yield and economic returns during <i>kharif</i> season. Groundnut varieties <i>Konkan Gaurav, TG 26</i> and <i>JL 776</i> are recommended to grow under organic package for getting higher yield and economic returns during <i>rabi</i> – hot weather season. (RARS., Karjat)	

65.	Sweet corn: Irrigation and nutrient management 1. In <i>Konkan</i> region on the newly developed terraced land, it is recommended to irrigate <i>rabi</i> sweet corn (var. <i>Madhu</i>) with 50 mm irrigation depth at 10 days interval for obtaining higher yield. 2. The fertilizer dose of 30 kg N (Urea) along with 18 tones of FYM per hectare is also recommended. (Dr. R.T. Thokal and Dr. T.N. Thorat)	2008			
66.	Cabbage: Pressurized irrigation and nutrient management 1. In <i>Konkan</i> region under lateritic soil it is recommended to irrigate <i>rabi</i> Cabbage (var. Golden Acre) by micro sprinkler irrigation with 13 mm irrigation at 3 days interval for obtaining higher yield. 2. The fertilizer dose of 120: 60: 60 kg NPK (Urea, SSP, MOP) ha⁻¹ is also recommended. (Dr. R.T. Thokal and Dr. T.N. Thorat)	2008			
67.	ECF project Due to ECF scheme employment had increased 21.31 per cent and income has increased by 84.13 per cent over non beneficiaries. It is recommended that for effective transfer of recommended agricultural technologies ECF scheme be implemented on cultivators field on large scale by the extension agencies and scientists of the University. (RARS., Karjat)				
68.	<i>Konkan Jalkund</i> - A micro rain water harvesting technique for horticulture crops on hill slopes of Konkan Relevance • Micro rainwater harvesting technology on hill slopes for newly planted mango and cashew grafts where farmers do not have access of water source. Key features of the technology ✓ Dimension: Pit size 4 x 1 x 1m or 2 x 1 x 2 m ✓ Capacity:4000 lits. per structure ✓ Lining & cushioning: HDPE, Silpaulin(200GSM) + paddy straw ✓ Silpaulin paper size: 7 x 4m or 7 x 6 m as per pit size ✓ Number of pits (ha): Mango-10 pits & Cashew-20 pits ✓ Percent survival of grafts: 85-87 % ✓ Construction cost: Rs.6,400/- ✓ Cost of rain water harvested: Rs.0.35-0.40 / lit. Monetary gains (0.5 ha area) <table> <tr> <td>District</td><td>Mango</td><td>Cashew</td></tr> </table>	District	Mango	Cashew	
District	Mango	Cashew			

	Ratnagiri Rs.41, 600/- Rs.30,800/- Sindhudurg Rs. 43,500/- Rs.32,900/- TSP programme Raigad & Palghar (Year 2013-14 to 2016-17) - Area enhancement under mango & cashew: 52.7 ha - Beneficiaries:264 - Water stored: 2108 meter cube Resolution for allocation of grants by Maharashtra Govt.: Rs.40 crores State Govt. grant allocations as subsidy : Rs. 2 crore 60 lakh (Dist.-Ratnagiri) and Rs. 2 crore 82 lakh (Dist- Sindhudurg)	
69.	Under lateritic soil conditions of Konkan region, it is recommended that the bitter gourd crop should be sown two weeks after onset of monsoon for getting highest fruit yield.	2005
70.	Under lateritic soil conditions of south Konkan region, it is recommended that the cucumber crop be sown immediately after onset of monsoon.	2005
71.	For effective and economical weed control in dolichos bean during rabi hot weather season two hand weedings one at 20 DAS and another at 45 DAS should be followed. However whenever man power is not available for manual operations like hoeing or hand weeding, pre emergence application of oxydiargyl @ 0.1 kg a.i. ha ⁻¹ be integrated given with hand weeding once at 45 DAS.	2005
72.	Agril. Meteorology In lateritic soil of Konkan region, bitter guard crop should be sown immediately after onset of monsoon (24 th MW, 11-17 June) for getting highest fruit yield.	2004
73.	For effective weed control in <i>rabi</i> chilli, fluchloralin @ 1 kg a.i. ha ⁻¹ may be applied as per plant incorporation followed by 1 hoeing one month after transplanting.	2004

d. Completed Research Projects/Programmes/Schemes

Title:

UR Nos.:

Objectives:

Name of PI/ Co-PI

Sponsoring

Agency:

Duration:

Total Outlay:

Summary of Achievements:

Relevant Photographs:

- e. **Ongoing Research Projects/Programmes/Schemes:** Only provide the name of the on going Research Projects/Programmes/Schemes. The details of the on going Research Projects/Programmes/Schemes will have to be provided by the concerned in charge in the separate format provided for this purpose. The link will be provided here with those details.

2. **Repository of abstracts of the theses:** Provide here the years wise details of the abstract of the theses/projects approved by the Department/Section for Bachelor, Masters and Doctoral theses in following format

Name of the candidate:
Degree for which the thesis/project report submitted:
Year of submission:
Name of the Guide/Co guide:
Abstract:

3. Extension Activities

a. The training programmes organized

Title:
Sponsorer:
Date and duration:
Participants: (Nature of the participation for eg. Farmers, Govt official, Academician etc and no. of participants)
Schedule of the training programme:
Special feature of the training programme: for eg training programme was especially for the women participant
One photograph

b. Seminar/Symposia/Conference/Workshop Organized

Title:
Sponsorer:
Date and duration:
Participants: (Nature of the participation and no. of participants)
Schedule of the Seminar/Symposia/Conference/Workshop:
Key Note Speakers along with their topic of speech
No. of papers presented
Whether papers published in abstract/full length form? If so provide the details in bibliographical format.
One photograph

c. Farmer Melawa Organized

Title:
Sponsorer:
Date and duration:
Participants: (Nature of the participation for eg. Farmers, Govt official, Academician etc and no. of participants)
Name of the speakers along with their topics
One photograph

d. Radio/TV Talks delivered by the staff members of the Department/Section:
Provide the relevant details such as name of the person, topic, where and when delivered etc.

e. Farmer-Scientist Forum: The name of the form along with the in charge of the forum, members of the forum (name, address and phone number) and activities of the forum be provided here.

f. Other Extension Activities: Provide the details of any other notable extension activities performed by the Department/Section

g. Publications: Provide the details of the following publications published by the Department/Section in bibliographical form

Books
Booklet/bulletin
Folders
Souvenir/Proceedings of Seminar/Symposia/Conference/Workshop
Organized
Training manuals of the training programme organized
Journal Research papers
Full length research papers published in Proceedings of Seminar/Symposia
/Conference/Workshop

4. Details of other activities (for e.g. seed production, production of other commodities etc)

5. Contact Information

Name of the Head
Name of the
Department Postal
Address
Landline
Number Mobile
Number Fax
Email

6. News and Events